COER UNIVERSITY

Lab Sheet 2: Data Manipulation Language (DML) Commands

Question 1

Question: Insert one record into the Student table.

Answer:

```
INSERT INTO Student (Student_ID, Name, Course, City, Fees)
VALUES (101, 'Aman Sharma', 'B.Tech CSE', 'Roorkee', 65000.00);
```

Explanation: Inserts a single row into the Student table specifying explicit values for all columns.

Question 2

Question: Insert five student records using separate INSERT statements.

Answer:

```
INSERT INTO Student VALUES (102, 'Riya Verma', 'B.Tech AI', 'Delhi', 70000.00);
INSERT INTO Student VALUES (103, 'Aarav Gupta', 'B.Tech CS', 'Meerut', 60000.00);
INSERT INTO Student VALUES (104, 'Sneha Kapoor', 'B.Tech DS', 'Delhi', 68000.00);
INSERT INTO Student VALUES (105, 'Rohan Mehta', 'B.Tech CSE', 'Roorkee', 65000.00);
INSERT INTO Student VALUES (106, 'Pooja Joshi', 'B.Tech AI', 'Meerut', 70000.00);
```

Explanation: Executes five individual INSERT statements to populate five distinct records into the table.

Question 3

Question: Insert multiple records into the Student table using a single statement.

Answer:

```
INSERT INTO Student (Student_ID, Name, Course, City, Fees) VALUES
(107, 'Karan Singh', 'B.Tech DS', 'Haridwar', 68000.00),
(108, 'Neha Sharma', 'B.Tech Cyber', 'Dehradun', 65000.00);
```

Explanation: Combines multiple value tuples in a single INSERT statement for multi-row insertion.

Question 4

Question: Insert records into the Employee table.

Answer:

```
INSERT INTO Employee (EmpID, Name, Department, Salary, City) VALUES
(1, 'Rohan Kumar', 'IT', 55000.00, 'Delhi'),
(2, 'Pooja Singh', 'HR', 48000.00, 'Meerut'),
(3, 'Amit Verma', 'Finance', 62000.00, 'Roorkee');
```

Explanation: Populates sample employee records into the Employee table.

Question 5

Question: Insert records into the Department table.

Answer:

```
INSERT INTO Department (DeptID, DeptName) VALUES
(1, 'Computer Science'),
(2, 'Information Technology'),
(3, 'Cyber Security');
```

Explanation: Populates department records into the Department table.

Question 6

Question: Insert records into the Product table.

Answer:

```
INSERT INTO Product (ProductID, ProductName, Price, Quantity) VALUES
(501, 'Laptop', 55000.00, 10),
(502, 'Mouse', 500.00, 0),
(503, 'Keyboard', 1200.00, 15);
```

Explanation: Populates product records into the Product table.

Question 7

Question: Insert records into the Customer table.

Answer:

```
INSERT INTO Customer (CustomerID, CustomerName, City) VALUES
(201, 'Amit Patel', 'Mumbai'),
(202, 'Suresh Kumar', 'Delhi');
```

Explanation: Populates initial records into the Customer table.

Question 8

Question: Insert records into the Course table.

Answer:

```
INSERT INTO Course (CourseID, CourseName, Fees) VALUES
(301, 'DBMS', 5000.00),
(302, 'Python', 4500.00),
(303, 'Web Dev', 6000.00);
```

Explanation: Inserts course details into the Course table.

Question 9

Question: Insert a student record with values for all columns.

Answer:

```
INSERT INTO Student VALUES (109, 'Vikas Gupta', 'B.Tech AIML', 'Delhi', 72000.00);
```

Explanation: When inserting values for every column in exact schema order, explicit column names can be omitted.

Question 10

Question: Insert a student record by specifying only selected columns.

Answer:

```
INSERT INTO Student (Student_ID, Name, Course)
VALUES (110, 'Ananya Roy', 'B.Tech Cyber');
```

Explanation: Specifies only mandatory columns (Student_ID, Name, Course), while unmentioned columns (City, Fees) take NULL.

Question 11

Question: Update the city of a particular student.

Answer:

```
UPDATE Student
SET City = 'Dehradun'
WHERE Student_ID = 101;
```

Explanation: Uses a WHERE clause to update the city specifically for Student_ID = 101.

Question 12

Question: Update the salary of a particular employee.

Answer:

```
UPDATE Employee
SET Salary = 60000.00
WHERE EmpID = 1;
```

Explanation: Modifies the salary for the employee with EmpID = 1.

Question 13

Question: Increase the salary of all employees by 10%.

Answer:

```
UPDATE Employee
SET Salary = Salary * 1.10;
```

Explanation: Omitting the WHERE clause applies a 10% raise across all records in the Employee table.

Question 14

Question: Increase the price of all products by 5%.

Answer:

```
UPDATE Product
SET Price = Price * 1.05;
```

Explanation: Applies a 5% price increase to all products in the Product table.

Question 15

Question: Change the course of a particular student.

Answer:

```
UPDATE Student
SET Course = 'B.Tech Data Science'
WHERE Student_ID = 102;
```

Explanation: Updates the course for Student_ID = 102.

Question 16

Question: Update the department of an employee.

Answer:

```
UPDATE Employee
SET Department = 'Finance'
WHERE EmpID = 2;
```

Explanation: Changes the department attribute for EmpID = 2.

Question 17

Question: Update the city of all students from Meerut to Delhi.

Answer:

```
UPDATE Student
SET City = 'Delhi'
WHERE City = 'Meerut';
```

Explanation: Updates City to 'Delhi' for all records where City currently equals 'Meerut'.

Question 18

Question: Update the fees of a particular course.

Answer:

```
UPDATE Course
SET Fees = 5500.00
WHERE CourseID = 301;
```

Explanation: Modifies the course fees for CourseID = 301.

Question 19

Question: Update multiple columns of a student simultaneously.

Answer:

```
UPDATE Student
SET Course = 'B.Tech AI&ML', City = 'Noida', Fees = 75000.00
WHERE Student_ID = 103;
```

Explanation: Updates Course, City, and Fees simultaneously in a single UPDATE statement.

Question 20

Question: Delete a particular student record using Student_ID.

Answer:

```
DELETE FROM Student
WHERE Student_ID = 110;
```

Explanation: Removes the student record corresponding to Student_ID = 110.

Question 21

Question: Delete all students belonging to a particular city.

Answer:

```
DELETE FROM Student
WHERE City = 'Meerut';
```

Explanation: Deletes all rows where City equals 'Meerut'.

Question 22

Question: Delete employees whose salary is below a specified amount.

Answer:

```
DELETE FROM Employee
WHERE Salary < 40000.00;
```

Explanation: Deletes employee records earning less than 40,000.

Question 23

Question: Delete products whose quantity is zero.

Answer:

```
DELETE FROM Product
WHERE Quantity = 0;
```

Explanation: Deletes out-of-stock products (Quantity = 0).

Question 24

Question: Delete records satisfying multiple conditions using AND/OR.

Answer:

```
DELETE FROM Student
WHERE City = 'Delhi' AND Fees < 65000.00;
```

Explanation: Deletes student records that satisfy both conditions simultaneously.

Question 25

Question: Perform a complete sequence of INSERT, UPDATE and DELETE operations on a Student table and display the table after each operation.

Answer:

```
Step 1: INSERT
INSERT INTO Student VALUES (199, 'Test Student', 'B.Tech', 'Roorkee', 50000.00);
SELECT * FROM Student WHERE Student_ID = 199;

Step 2: UPDATE
UPDATE Student SET Fees = 52000.00 WHERE Student_ID = 199;
SELECT * FROM Student WHERE Student_ID = 199;

Step 3: DELETE
DELETE FROM Student WHERE Student_ID = 199;
SELECT * FROM Student WHERE Student_ID = 199;
```

Explanation: Demonstrates a full CRUD sequence (Create, Read, Update, Delete) on a single record, checking table state after each step.